of a given string theory. And to the surprise of most everyone in the field, the strong coupling properties of the Heterotic-O string appear to be identical to the weak coupling properties of the Type I string, and vice versa. Moreover, the strong coupling physics of the Type IIB string is identical to its own properties when its coupling is weak. These unexpected links encourage us to follow Witten and press on to the other two string theories, Type IIA and Heterotic-E, to see how they fit into the overall picture. Here we will find even more exotic surprises. To prepare ourselves, we need a brief historical digression.

Supergravity

In the late 1970s and early 1980s, before the surge of interest in string theory, many theoretical physicists sought a unified theory of quantum mechanics, gravity, and the other forces in the framework of point-particle quantum field theory. The hope was that the inconsistencies between point-particle theories involving gravity and quantum mechanics would be overcome by studying theories with a great deal of symmetry. In 1976 Daniel Freedman, Sergio Ferrara, and Peter Van Nieuwenhuizen, all then of the State University of New York at Stony Brook, discovered that the most promising were those involving supersymmetry, since the tendency of bosons and fermions to give cancelling quantum fluctuations helps to calm the violent microscopic frenzy. The authors coined the term supergravity to describe supersymmetric quantum field theories that try to incorporate general relativity. Such attempts to merge general relativity with quantum mechanics ultimately met with failure. Nevertheless, as mentioned in Chapter 8, there was a prescient lesson to be learned from these investigations, one that presaged the development of string theory.

The lesson, which perhaps became most clear through the work of Eugene Cremmer, Bernard Julia, and Scherk, all of the Ecole Normale Supérieure in 1978, was that the attempts that came closest to success were supergravity theories formulated not in four dimensions, but in more. Specifically, the most promising were the versions calling for ten or eleven dimensions, with eleven dimensions, it turns out, being the maximal possible.11 Contact with four observed dimensions was accomplished in the framework, once again, of Kaluza and Klein: The extra dimensions were curled up. In the ten-dimensional theories, as in